

Freight Class Calculator

Implementation summary — MSW Overwatch

MetroShoe Warehouse
September 2026 · App v3.4.0

STATUS: SUCCESSFULLY IMPLEMENTED

Freight Class Calculator is a gated LTL tool under Tools → Freight Class. Authorized users enter pallet dims manually or bulk-upload Excel, get NMFC density class (XPO-compatible, including the optional 75" height rule), then copy or export a Summary workbook.

WHAT IT DOES

Freight Class Calculator turns pallet dimensions and weights into an NMFC density-based LTL freight class (13-sub scale, effective July 2025). Staff can calculate one shipment by hand or many shipments from Excel, then share results without leaving Overwatch.

BUSINESS VALUE

- One consistent XPO-compatible class for warehouse / ops quotes
- Bulk Excel avoids re-keying multi-pallet FBA / vendor shipments
- Optional 75" rule matches carrier behavior when needed
- Gated to named warehouse users + superadmin

HOW A CALCULATION WORKS

Step	What happens
1. Open tool	Sidebar Tools → Freight Class (allowlisted + superadmin only)
2. Choose mode	Manual entry or Bulk upload of the Excel template
3. Height rule	Optional toggle: skip XPO 75–95" → 96" bump before density
4. Calculate	Rows grouped by Shipment ID; density (lb/ft³) maps to NMFC 13-sub class
5. Share	Copy summary, or export Summary + Line Items workbook

MODES & OUTPUTS

KEY RULES

Mode	Input → Output	Pallet Number ≠ Pallet Count — Number is a label; Count multiplies volume Density = total weight ÷ total cubic feet (after height rule) Legacy column name Pallets still maps to Pallet Count Audit: calculate / export / template download events
Manual	Pallet rows in UI → class + density on screen	
Bulk upload	.xlsx template → multi-shipment results	
Export	Results → Freight Class Summary.xlsx	
Copy	Tab-separated summary to clipboard	

EXCEL TEMPLATE COLUMNS

Sheet name: Pallet Dims. Download from the tool or API /freight-class-calculator/template.

Column	Meaning
Shipment ID	Groups pallet rows (blank rows inherit prior ID)
Pallet Number	Line label only (#1, #2, ...) — not used in math
Pallet Count	How many identical pallets this row represents (volume multiplier)
Weight	Total weight (lbs) for that row
Length / Width / Height	Inches; height may be adjusted by 75" rule

WHO HAS ACCESS

NMFC DENSITY SCALE

User	Access	Density (lb/ft³)	Class
Allowlisted emails	sunshine@, stephanie@, paulo@, paulo@, johnbernard@ (metroshoewarehouse.com)	50+	50
		35 – 50	55
		30 – 35	60
Superadmins	Full Freight Class access	22.5 – 30	65
All other users	No sidebar link; route blocked by API + UI gate	15 – 22.5	70
		12 – 15	85
		10 – 12	92.5
		8 – 10	100
		6 – 8	125
		4 – 6	175
		2 – 4	250
		1 – 2	300
		< 1	400

ROLLOUT CHECKLIST

- Confirm Freight Class appears under Tools only for allowlisted emails
- Download template; verify headers: Shipment ID, Pallet Number, Pallet Count, Weight, L/W/H
- Bulk-upload the sample rows; confirm two shipments and correct classes
- Toggle Skip 75" height rule on a 80" pallet and compare density

- Manual mode: enter Pallet Count (not Number); Calculate; Copy summary
- Export Freight Class Summary.xlsx (Summary + Line Items with #n labels)
- Legacy file with old Pallets column still calculates
- Non-allowlisted user: no sidebar entry; API returns restricted

NOTES FOR LEADERSHIP

Shipped on web app version 3.4.0 (backend service + React Tools page)

Route: /freight-class-calculator · Sidebar: Tools → Freight Class

Allowlist lives in backend config + matching frontend freightClassAccess.ts (keep in sync)

Does not touch Keepa jobs, Daily Runs, MAP, or warehouse catalog data

Class is density-only guidance; carrier quotes may still apply additional NMFC item / packaging rules